

Boundary Management and Work-Nonwork Balance While Working from Home

Tammy D. Allen* 

University of South Florida, USA

Kelsey Merlo

University of South Florida, USA

Roxanne C. Lawrence

University of South Florida, USA

Jeremiah Slutsky

University of South Florida, USA

Cheryl E. Gray

University of South Florida, USA

Within the wake of the COVID-19 pandemic, we investigate work-nonwork boundary management among workers who transitioned to remote work. Based on five waves of data and a sample of 155 remote workers, we find that the preference for segmentation was associated with greater work-nonwork balance. We also found that having a dedicated office space within the home and fewer household members was associated with greater work-nonwork balance. However, these variables did not moderate the relationship between segmentation preferences and work-nonwork balance as expected. We discuss implications for future research on boundary management processes and practices.

INTRODUCTION

COVID-19 has intertwined work and nonwork roles in an unprecedented way. Millions of workers have literally brought their work home with them, transforming the home into both home and office (Koetsier, 2020). This situation

* Address for correspondence: Tammy D. Allen, Department of Psychology, University of South Florida, 4202 E. Fowler Ave., PCD 4118G, Tampa, FL 33620, USA. Email: tallen@usf.edu

sets the stage for role boundary management and work-nonwork balance challenges as newly remote workers are confronted with blurred work and nonwork role boundaries. Moreover, sudden remote work can make it hard for individuals to enact their preferred boundary management strategies, especially workers who prefer segmentation.

Against the backdrop of the ongoing COVID-19 global health crisis, the need to increase our understanding of relationships between boundary management and work-nonwork issues is more pressing than ever. Thus, the objective of the current study is to advance our understanding of boundary management among a sample of workers who transitioned to remote work due to COVID-19. Specifically, guided by boundary theory, we investigate the relationship between integration-segmentation preferences and work-nonwork balance across a three-month period. In addition, we test several contextual moderators associated with the home environment (i.e., number of other individuals in the home, having a dedicated office with a door) thought to moderate this relationship. Finally, through open-ended question responses we identify the strategies implemented by remote workers to create work-nonwork boundaries.

In investigating these issues, we make several contributions to the existing literature. Research interest in remote work (or telecommuting), as well as on boundary management, has grown steadily in recent years (e.g., Allen, Cho, & Meier, 2014; Allen, Golden, & Shockley, 2015; Kossek, Lautsch, & Eaton, 2006; Lapierre, Van Steenbergen, Peeters, & Kluwer, 2016). The extant remote work literature generally compares samples of remote and non-remote employees to learn about the outcomes of remote work (e.g., Gajendran, Harrison & Delaney-Klinger, 2015; Martin & MacDonnell, 2012). Providing the flexibility to work remotely is thought to enable work-nonwork balance among workers who opt for such an arrangement. However, remote work is typically a voluntary option that workers choose for themselves (Lapierre et al., 2016). Because many individuals have been forced into remote work as a result of COVID-19, the current context presents a unique opportunity for examining some of the tenets associated with boundary management theory under conditions in which self-selection effects are mitigated.

In addition, by investigating work-nonwork balance, we expand the existing literature that examines interrole outcomes in association with boundary management beyond that of work-family conflict. This is in line with calls for research to expand our understanding of the connections between work and the myriad of other roles in life that go beyond family within contemporary society (Powell, Greenhaus, Allen, & Johnson, 2019; Rothausen, 2011). Work-nonwork balance is a key outcome for examination in the context of the pandemic as work demands (e.g., use and adaptation to new virtual work tools) and nonwork demands (e.g., lack of dependent care) have shifted

and increased for many workers (e.g., Craig & Churchill, 2020). Moreover, our examination of work-nonwork balance based on five waves of data is a unique and valuable contribution as there is little data available concerning how stable or how dynamic work-nonwork balance is across time (Casper, Vaziri, Wayne, DeHauw, & Greenhaus, 2018).

Another contribution of our study is a consideration of boundary management supplies within the home environment. Existing research has tended to examine features of the work environment (e.g., flexible scheduling, onsite childcare) in relation to boundary management (e.g., Rothbard, Phillips, & Dumas, 2005). However, features of the home environment are likely to play a pivotal role in how well remote workers are able to enact their boundary management preferences and manage their work and nonwork roles. Accordingly, our incorporation of contextual moderators generates new knowledge applicable to the remote workforce. We further contribute to the boundary management and the remote work literatures by investigating the specific boundary management tactics that individuals report using while working from home. Such information can help provide insights that may inform practical guidance that can be offered to remote workers.

LITERATURE REVIEW

The past decades have been marked by a rapid increase in the availability of technological tools that have changed where we work and the way by which work is done (Olson-Buchanan, Boswell, & Morgan, 2016). These new technologies have enabled an increase in the number of individuals who work remotely (US Bureau of Labor Statistics, 2018). Even so, prior to COVID-19, remote work was an option for only a small percentage of the workforce (Desilver, 2020). Never before has society witnessed a situation in which millions of workers have been forced to transition to remote work across the globe. Although working from home may appeal to some, the diminished boundary between work and nonwork comes with challenges as well.

Boundary Management

Boundary management concerns the ways that individuals create, maintain, or change boundaries in order to effectively navigate the world around them, including their work and nonwork roles (Ashforth, Kreiner, & Fugate, 2000; Nippert-Eng, 1996). The differentiation of space and time traditionally serves as a device that defines different roles. For example, the role of worker is commonly enacted at a place of work and during specified hours (e.g., Monday–Friday, 9–5) while nonwork roles are typically enacted while one is physically located outside of the office during the evening and weekends (Allen et al., 2014). As Ashforth et al. (2000) explain, role boundaries can vary from highly

segmented, in which each role has a strict location and time, to highly integrated, in which multiple roles can occur within the same location and time. Segmentation/integration can be considered in terms of personal preferences, actual behaviors enacted by individuals, and the environmental conditions that facilitate segmentation/integration (e.g., boundary supplies) (Kreiner, 2006; Wepfer, Allen, Brauchli, Jenny, & Bauer, 2018). The juxtaposition of integration and segmentation are viewed as lying along a continuum with most individuals and situations falling somewhere in between the two extremes (Bulger, Matthews, & Hoffman, 2007; Nippert-Eng, 1996).

Individuals with a preference toward integration are comfortable removing boundaries between work and nonwork, while those with a preference toward segmentation prefer to keep temporal and physical boundaries between work and nonwork intact (Ashforth et al., 2000). Boundary management preferences are important to the extent that individuals are able to act in ways consistent with their needs and preferences. Typically, individuals can enact behavioral strategies and create contextual conditions that facilitate their ability to manage boundaries in a way that aligns with their personal preferences (Kossek, 2016; Kossek & Lautsch, 2012; Kreiner, Hollensbe, & Sheep, 2009). For example, those who prefer segmentation may be less likely to opt for remote work arrangements because it is incongruent with their desire to segment life roles. There is some support for this supposition. Specifically, Shockley and Allen (2010) found a negative relationship between preference for segmentation of work from nonwork roles and the extent individuals who could work remotely (i.e., university faculty) did indeed work remotely.

Boundary Management Preferences and Work-Nonwork Balance

The ability to achieve a sense of balance across life roles is often considered an elusive, but desirable state of being (Kreiner et al., 2009). The advent of the COVID-19 pandemic, coupled with the rapid transition to remote work, has brought the challenges associated with work-nonwork balance into sharp relief. Work-nonwork balance reflects an overall appraisal of one's effectiveness and satisfaction with work and nonwork roles (Greenhaus & Allen, 2011). It is theorized that individuals are more likely to experience poor work-family outcomes if there is misalignment between their boundary management preferences and the degree to which they are able to satisfy that preference. A small body of research exists that has investigated the relationship between boundary management preferences and work-nonwork variables, primarily work-family conflict (e.g., Chen, Powell, & Greenhaus, 2009; Kreiner, 2006; Shockley & Allen, 2010). Results of these studies provide little support for a direct relationship between boundary management

preferences and work-family conflict. However, as noted by Kreiner (2006), neither integration nor segmentation of roles is inherently good or bad. What is important is fit between the individual and what the context provides.

Boundary theory suggests that individuals are likely to feel more effective in their work and nonwork roles when individual boundary management preferences are aligned with their environment (Chen et al., 2009; Cho, 2020; Kreiner, 2006; Rothbard et al., 2005). Remote work alters the traditional boundaries between work and nonwork domains. Because segmentors prefer to keep work and nonwork roles separate, a working situation that reduces boundaries such as working from home, would ostensibly be incongruent with their preferences and result in misalignment (Rothbard et al., 2005). This inconsistency may make it difficult to strike a balance between work and nonwork. Moreover, the weak role boundaries associated with remote work should be more disruptive to those who prefer segmentation. By contrast, those who prefer integration of work and nonwork roles should more readily be able to adapt and thrive in the remote work from home environment. Some evidence for the greater difficulty associated with COVID-19 is supported by Vaziri, Casper, Wayne, and Matthews (2020) who studied transitions in workers' work-family interfaces (i.e., conflict and enrichment) before and after the onset of COVID-19. Based on latent profile analyses, they found that employees with lower segmentation preferences were more likely to benefit during COVID-19 by transitioning from an active (medium conflict and enrichment) to a beneficial (low conflict and high enrichment) profile. The researchers suggest that individuals who prefer segmentation are more likely to face adverse changes to the work-family interface. Even those who prefer integration may find that the novelty of complete overlap is difficult (Shockley & Clark, 2020), but we expect that those with segmentation preferences will be most challenged with regard to work-nonwork balance.

Hypothesis 1: Individuals who prefer segmentation of work and nonwork roles report less work-nonwork balance than individuals who prefer integration.

We also expect that the relationship between boundary management preferences and work-nonwork balance will change in slope across time as individuals have the opportunity to adjust to their new remote work situation. We make our prediction on the basis of adaptation theory. Adaptation theory suggests that in the short term, stressors or shock events will have a negative impact on individual well-being. However, over a longer period of time, individuals adapt to their new situation and well-being returns (Brickman, Coates, & Janoff-Bulman, 1978). Although the rate of adaption varies across life events and as a function of individual differences, the adaptation pattern has been observed in response to a wide variety of stressors (Diener, Lucas, &

Scollon, 2006; Luhmann, Hofmann, Eid, & Lucas, 2012). Thus, while we expect a negative relationship between segmentation preferences and work-nonwork balance, we propose that time alters the magnitude of the relationship as individuals adapt to remote work.

Specifically, the strength of the relationship between boundary management preferences and work-nonwork balance is expected to diminish across time.

Hypothesis 2: The negative relationship between segmentation preferences and work-nonwork balance diminishes across time.

Contextual Moderators

We consider several factors that may enhance or diminish an individual's ability to segment work and nonwork while working from home. As individuals vary in their preferences for integration/segmentation, an individual's environment also varies with regard to the extent that it promotes integration/segmentation (Kreiner, 2006). That is, workplaces vary with regard to the extent that they supply conditions that facilitate integration/segmentation. These differences are important in that workplaces that enable individuals to segment work and nonwork roles (i.e., employees are not expected to take calls after work hours) have been associated with less work-family conflict (Chen et al., 2009; Kreiner, 2006).

The fit between person and environment has long been considered important to individual well-being (Edwards, Cable, Williamson, Lambert, & Shipp, 2006; Edwards & Rothbard, 1999; Kristoff, 1996). A comprehensive discussion of the various theoretical person-environment fit approaches is beyond the scope of this paper. However, one stream of fit research is primarily applicable to the current study. *Needs-supplies* fit refers to the match between an individual's psychological needs and/or values and what the environment supplies to fulfill those needs. Within the boundary management literature, research has focused on a form of needs-supplies fit that considers integration/segmentation preferences as a need and what the workplace environment offers as a supply to that need (Kreiner, 2006). Workplace supplies in this case can refer to objective social and physical features of the environment, such as having onsite childcare (an integration supply), as well as subjective perceptions about the environment, such as the extent that the workplace lets workers forget about work when they are at home (a segmentation supply) (Chen et al., 2009; Edwards & Rothbard, 1999; Kreiner, 2006).

The focus thus far within the boundary management literature has been on the supplies associated with the office or workplace environment. Research has yet to consider the boundary management physical and social features

(i.e., supplies) within the home environment that may promote integration or segmentation, which seem a crucial consideration for those who work from home. Accordingly, we consider the availability of a dedicated home office as a physical boundary management supply that promotes segmentation and the number of others within the home as a social boundary management supply that promotes integration (thus, making segmentation more difficult).

Common boundary management advice to remote workers is to establish a dedicated workspace within the home (e.g., Shockley & Clark, 2020). A home office set up is likely more common for those who plan remote work. However, many who have been thrust into remote work due to the pandemic may lack such a luxury. The ability to erect a physical boundary, such as by having a dedicated home office, should facilitate segmentation. Moreover, a home office can enhance privacy and give one the psychological sense of “being at the office.” Lack of a home office can exacerbate the unmet needs of those who prefer segmentation, further reducing work-nonwork balance. Thus, we predict that a home office can strengthen the relationship between segmentation preferences and work-nonwork balance.

Hypothesis 3: The negative relationship between segmentation preferences and work-nonwork balance is stronger among those without a dedicated home office than among those with a dedicated home office.

The second boundary management supply we consider is the number of other individuals in the home. Those living alone should have an easier time establishing boundaries than those who are living with others, thus benefiting those with segmentation preferences. An integrating supply, such as sharing both a workplace and a home with the same people, can be taxing to those who prefer segmentation. Family members and roommates can increase noise in the home, interrupt workers with nonwork matters while they are working, and be a consistent reminder that the worker is at home rather than at the workplace. Moreover, the more household members who share the home with the worker, the more challenging it can be for workers who prefer segmentation to negotiate a firm work-nonwork boundary that meets their own needs.

Hypothesis 4: The negative relationship between segmentation preferences and work-nonwork balance is stronger among those with more members in the household than among those with fewer members in the household.

Boundary Management Tactics

Thus far, we have hypothesized relationships involving boundary management preferences and boundary management supplies in relation to work-nonwork

balance. The rapid transition to remote work also presented an opportunity wanted to gain insight into the tactics that individuals were using to boundary manage while working from home. Specifically, through an open-ended response qualitative coding approach, we investigated the strategies individuals implemented to segment their work and nonwork roles among those who were still working primarily remotely in August 2020. Despite the growth of the work-home boundary literature (Allen et al., 2014), relatively little attention has been paid to the specific tactics that individuals use to create and maintain preferred boundaries. Kreiner et al.'s (2009) work delineated specific tactics individuals could use in response to conflicting demands in the workplace and at home. The current research provided an opportunity to examine segmenting tactics in the context of remote work.

Tactics can be thought of as the decisions or actions that individuals take to recalibrate the work-home boundary. Based on qualitative work with priests, Kreiner et al. (2009) identified four categories of boundary management tactics. Behavioral tactics involve using other people, such as getting help from others. Temporal tactics involve controlling work time, such as by blocking off segments of time for work and nonwork. Physical tactics refer to adapting physical boundaries, such as by manipulating physical space. Finally, communication tactics refer to setting expectations, such as by confronting those who violate boundaries. In the current study, we investigate the strategies remote workers report implementing to segment work and nonwork.

Research Question 1: What strategies do individuals use to segment work and nonwork when working from home?

METHOD

Participants

Participants were 155 working adults recruited through Prolific, an online research platform. Only participants who were working full-time (30+ hours/week), reported limited prior experience working remotely (less than one day/week), and spent a minimum of 75 percent of their time working from home during all waves of the study were included. Participants worked in a variety of industries and held an array of job titles such as teacher, engineer, IT director, sales manager, and administrative assistant. At Wave 1, a total of 317 individuals participated. Across study waves, a total of 58 participants dropped out of the study, 16 participants were removed for being ineligible for the study (e.g., were not working from home due to COVID-19 or frequently worked from home prior to COVID-19), 19 participants were removed because they

failed to respond appropriately to attention/comprehension checks, 4 were removed for reporting a change in full-time employment status, 4 were removed for technical issues, and 61 were removed because they began working from home less than 75 percent of the time. The majority of the sample were male (59%), White (85%), unmarried (54%), and without children (63%). The average age of the sample was 37 years ($SD = 9.28$).

Procedure

To obtain a sample of participants who transitioned to remote work due to COVID-19, two screening questions were employed. The first was, "Are you currently working from home full-time due to COVID-19?" Participants had to respond "yes" to continue. The second question was, "Prior to the social distancing policies associated with COVID-19, how often did you work from home (e.g., completed work tasks at home during normal business hours)?" Response options were: "never," "a few times a year or less," "once a month or less," "a few times a month," "once a week," "a few times a week," "everyday." Only those who responded once a week or less were able to continue in the study.

Changes in work-nonwork balance were captured using a longitudinal panel design. The first four waves of data were collected on a bi-weekly basis over a period of eight weeks (four waves) starting on May 13, 2020. Finally, a fifth wave of data was collected on August 3, 2020, six weeks after Wave 4. For each wave of the study, participants reported their work-nonwork balance. Segmentation preferences, designated office space, and number of household members were measured in the first wave of the study. Segmentation strategy responses were collected in the fifth wave for those who were working exclusively from home.

Measures

Segmentation Preferences. Six items based on Kreiner's (2006) segmentation preferences scale were used (e.g., "I don't like to have to think about work while I'm at home"; "I prefer to keep my non-work life at home"). Responses were made on a 5-point scale that ranged from "strongly disagree" to "strongly agree." Higher scores indicate greater preferences for segmentation of work and nonwork roles. Segmentation preferences were measured at Time 1. Coefficient alpha = .83.

Work-Nonwork Balance. Three items modified from the Greenhaus et al. (2012) work-family balance measure (e.g., "I am able to balance the demands of my work and nonwork") were used. Responses were made on a 5-point scale that ranged from "strongly disagree" to "strongly agree."

Higher scores indicate greater balance. Work-nonwork balance was measured at all five timepoints. Reliability across measurement occasions (R_{kf}) = .98; generalizability of change (R_c) = .80 (Shrout & Lane, 2012).

Segmentation Strategies. At Time 5, participants who were exclusively working from home responded to questions with regard to boundary management. The first question was, “Since transitioning to full-time remote work from home, have you implemented any strategies to create a boundary between work and nonwork?” Response options were “no” and “yes.” Those who responded “yes” were given a second prompt, “Please describe the strategies that you are using to create a boundary between work and nonwork.”

Household Characteristics. Additional single item measures of household characteristics were included. Specifically, participants were asked to report their household size (“Please indicate the number of persons currently living in your household (including yourself)”) and if they had a dedicated home office (“yes” or “no”).

RESULTS

Preliminary Analyses

Independent sample t -tests were conducted to determine if the sample of participants who remained in the study differed from the sample of participants who were removed from the study (either due to attrition, failed attention checks, or change in working status). Several significant differences were found between participants who remained in the study and participants who were removed from the study. First, participants who remained in the study were older, $t(309) = -2.08$, $p = .038$, and more educated $t(309) = -2.55$, $p = .011$, than those who were removed from the study. While these differences are statistically significant, the effect sizes for these differences are small. Cohen’s D for each of these differences is less than .5 (Cohen’s D for age = .24, Cohen’s D for education = .29). There was no statistical difference between participants who remained in the study versus participants who were removed from the study on any of the following variables: sex, household size, or boundary management preference at Wave 1.

Significant group differences were found for having a dedicated home office and work-nonwork balance at Wave 1. In particular, individuals who remained in the study were more likely to have a dedicated home office and reported higher work-nonwork balance than were those who were removed from the study, $t(309) = 2.17$, $p = .03$ and $t(309) = -1.99$, $p = .048$, respectively.

TABLE 1
Descriptive Statistics

	<i>M</i>	<i>SD</i>	1	2	3	4	5	6	7	8
1. Segmentation preferences	4.41	0.59	(.83)							
2. Household size	2.68	1.23	.04	-						
3. Home office ^a	0.43	0.50	-.09	.09	-					
4. Work-nonwork balance T0 ^b	4.02	0.86	.14	-.16*	.15	(.88)				
5. Work-nonwork balance T1 ^b	4.05	0.82	.21*	-.18*	.11	.75**	(.87)			
6. Work-nonwork balance T2 ^b	4.06	0.83	.23**	-.11	.15	.63**	.71**	(.88)		
7. Work-nonwork balance T3 ^b	4.02	0.95	.26**	-.07	.16	.60**	.72**	.71**	(.93)	
8. Work-nonwork balance T6 ^b	4.03	1.01	.21**	-.10	.16*	.57**	.59**	.61**	.55**	(.95)

Note. *N* = 155.
^a0 = no dedicated home office; 1 = dedicated home office.
^bTime was recoded to maintain linearity (e.g., Wave 1 = Time 0, Wave 2 = Time 1, Wave 3 = Time 2, Wave 4 = Time 3, Wave 5 = Time 6).
**p* < .05;
***p* < .01.

Post hoc analyses with participants who were working remotely and on-site ($N = 216$) indicated a significant effect of having a dedicated home office on work-from-home status. Specifically, individuals with a home office were more likely to be working from home at Wave 5, $b = -.64$, $SE = .25$, $p = .01$.

Hypothesis Testing

Table 1 provides descriptive statistics and intercorrelations for the study variables. To account for the nested structure of the data (e.g., observations within person), multilevel analyses were used to test Hypotheses 1–4. To model the changes in work-nonwork balance over time, time was included as an independent variable. Segmentation preferences was a grand-mean centered Level 2 variable. When testing Hypothesis 2, time was coded so that 0 represented our first assessment. As the response interval changed between Wave 4 and Wave 5 (e.g., from two weeks to six weeks), time was successively coded at subsequent waves such that each one-unit increase in the time variable corresponded to a two-week increase in time passing (i.e., Wave 1 = Time 0, Wave 2 = Time 1, Wave 3 = Time 2, Wave 4 = Time 3, Wave 5 = Time 6).

Hypothesis 1 stated that individuals who prefer segmentation of work and nonwork roles would report less work-nonwork balance than individuals who prefer integration. First, a multilevel model was conducted to determine if individuals with segmentation preferences reported less work-nonwork balance, on average, compared to those with integration preferences. A significant effect of boundary management preferences was found on average levels of work-nonwork balance ($b = .32$, $SE = .10$, $p = .002$). However, it was the opposite direction of the hypothesized effect. Individuals who expressed segmentation preferences exhibited higher work-nonwork balance than those expressing integration preferences (Hypothesis 1, not supported). Hypothesis 2 stated that the negative relationship between segmentation preferences and work-nonwork balance would diminish across time. Time did not moderate the relationship between boundary management preferences and work-nonwork balance ($b = .03$, $SE = .02$, $p = .17$) (Hypothesis 2, not supported).

As an exploratory analysis, we also tested for curvilinear effects, but none were detected. Specifically, the curvilinear effect of time on work-nonwork balance was nonsignificant, $b = -.001$, $SE = .005$, $p = .84$. Furthermore, the interaction between curvilinear time and boundary management preferences on work-nonwork balance was nonsignificant, $b = .002$, $SE = .003$, $p = .44$.

Hypothesis 3 stated that the negative relationship between segmentation preferences and work-nonwork balance would be stronger among those without a dedicated home office than among those with a dedicated home office. Having a dedicated home office space was related to significantly higher levels of work-nonwork balance ($b = .30$, $SE = .12$, $p = .01$). However, having

a home office space did not moderate the relationship between segmentation preferences and work-nonwork balance ($b = .14$, $SE = .20$, $p = .49$) (Hypothesis 3, not supported).

Hypothesis 4 stated that the negative relationship between segmentation preferences and work-nonwork balance would be stronger among those with more members in the household than among those with fewer members in the household. Having fewer people in the home was related to higher levels of work-nonwork balance ($b = -.10$, $SE = .05$, $p = .0495$). However, the number of people in the household did not moderate the relationship between boundary management preferences and work-nonwork balance ($b = -.004$, $SE = .09$, $p = .96$) (Hypothesis 4, not supported).

Research Questions

Research Question 1 concerned the strategies individuals reported that they used to segment work and nonwork when working from home. To address this question, we asked participants who were working exclusively from home at Wave 5 of the study to indicate if they had implemented a segmentation strategy. At Wave 5, 119 participants indicated that they were working from home 100 percent of the time. Of those 119 participants, 67 indicated that they had implemented a segmenting strategy. Responses to our open-ended question were content analyzed. We used the subcategories identified by Kreiner et al. (2009) as a base, and also permitted additional (sub)categories to emerge. We also allowed for adjustments to the definition of the tactics defined by Kreiner et al. (2009) to match the remote work context.

Our content coding process was as follows. First, one author went through all of the comments and placed them into an existing subcategory or into a new category. Each response could include multiple strategies and were coded as such. Second, the new categories were reviewed and discussed with a second author. The two authors agreed that three additional categories emerged that were not identified by Kreiner et al (2009). Third, three coders independent to the study repeated the coding process. The coders were provided with the definitions for each of the categories and asked to place each strategy (multiple strategies could be identified by each participant response) within one of the existing categories or to indicate that the response did not fit into any of the existing categories. We calculated the interrater reliability among the three coders using Randolph's (2008) free-marginal multirater kappa. Raters reached an overall 86.89 percent agreement ($kappa = .84$, 95% CI [.77, .91] (Fleiss, 1981 as cited in Randolph, 2008).

Table 2 shows each of the categories for segmentation strategies, a brief definition (complete definition provided to coders available upon request), the number of responses coded into that category and sample responses. Across

TABLE 2
Qualitative Coding of Segmentation Strategies

Category	Definition	Number of Comments	Sample Comments
Behavioral Tactics			
Using other people	Utilizing the skills and availability of other individuals	0	
Leveraging technology	Using technology to facilitate boundary work	2	"I created an alarm that warns me if I should have stopped working an hour before." "I'm setting timers to remind me to finish."
Invoking triage	Prioritizing seemingly urgent and important work and home demands	0	
Allowing differential permeability	Choosing which specific aspects of work-home life will or will not be permeable. Deciding which parts of the home and work to keep overlapped	0	
Emulating office routine*	Recreating the on-site office environment, routines, and overall feel of "going to work"	4	"Go outside after work and then come home so I feel like I'm coming home from the office." "I make sure I get dressed in 'real clothes' when I'm working, and then at the end of the workday, I go put my pajamas back on!" "I try to set a start time and a finish time for work so that I am able to work those hours and then stop working and enjoy my leisure hours." "I set strict working times and stop for the day at the designated time."
Temporal Tactics			
Controlling work time	Manipulations of one's regular or sporadic plans. Managing the time one <i>does</i> work	31	

(Continues)

TABLE 2
(Continued)

Category	Definition	Number of Comments	Sample Comments
Finding respite	Removing oneself from work-home demands for a short or significant amount of time (e.g., breaks, vacations)	6	"For lunchtime I try to get away from my workstation by going for a walk so that I am less likely to be thinking about work." "Taking lunch breaks"
Purposefully disconnecting*	Actively doing things to take the mind off work. Managing the time one <i>does</i> not work	22	"The only strategy I use is to disconnect completely when I finish my work; I silence work groups on my smartphone and try not to look at the messages." "As soon as I finish work, I store my laptop and work essentials, so I can't see them when I am off work."
Reducing work and home role overlap*	Working while family members are not around or asleep to avoid distractions	2	"This prevents interruptions that may occur during the course of the day to not have as great an impact as it might if I was trying to make sure I got all my work done within a tight timeframe." "Doing housework before start work so not distracted by it when working."
Physical Tactics			
Adapting physical and psychological boundaries	Erecting or dismantling physical or psychological borders or barriers between work and home domains.	9	"I work from the kitchen so do not go into the living room during working hours to keep that separate." "I also have bought a desk and have my own set up, rather than working from the dining table."
Manipulating physical space	Creating or reducing a physical distance or "no man's land" between the work and home domains.	27	"I have an office dedicated to working in, that I do not use at all when I am not working." "I have been using one bedroom exclusively as a study. I have avoided doing work in any room other than this room where possible."

(Continues)

TABLE 2
(Continued)

Category	Definition	Number of Comments	Sample Comments
Managing physical artifacts	Using tangible items such as work computers and phones, calendars, keys, photos, and mail to separate or blend aspects of each domain	5	"No other devices are taken into the [spare bedroom] so no other distractions are there." "I have a separate laptop for work and non-work."
Communicative Tactics			
Setting expectations	Managing expectations in advance of a work-home boundary violation	4	"Informing household members that I am unavailable during working hours." "I have agreed with my family that on certain hours I cannot be disturbed (unless necessary) and cannot have people enter my work room."
Confronting violators	Telling violator(s) of work-home boundaries either during or after a boundary violation	0	

* New subcategory.

the sample, 10 different boundary management strategies were identified. Of the boundary management tactics defined by Kreiner et al. (2009), temporal tactics were the most commonly reported. Specifically, many individuals stated that they strictly defined and controlled the time they started and ended work (*Controlling work time*) as well as when they took breaks (*Finding respite*). Physical tactics were the second most employed. Traditionally, workers were able to create geographical distance and barriers between work and nonwork by living away from their workplace (Kreiner et al., 2009). To create physical distance between work and nonwork domains in the work from home environment, individuals used dedicated home offices that allowed them to “go to” work at the start of the workday and “leave” work at the end of the workday (*Manipulating physical space*).

Individuals also created physical and psychological boundaries between their work and nonwork domains by declaring specific locations in their home (e.g. their kitchen) a designated workstation (*Adapting physical and psychological boundaries*). This tactic differed from manipulating physical space because it required a psychological reframing of one’s workstation from work to nonwork at the end of the day (e.g., the kitchen transitioned from a workstation to a traditional kitchen after work hours). Conversely, a dedicated home office allowed for a physical movement from work to nonwork; one could leave work behind by exiting the home office. The final physical tactic was *Managing physical artifacts* (i.e., tangible items that symbolize the work or home domain) to prevent the cross-contamination of work and nonwork. For example, individuals described keeping all work devices in their dedicated office and keeping separate laptops for work and nonwork.

In addition to temporal and physical tactics, individuals working from home applied behavioral tactics such as *leveraging technology* to create and maintain work-nonwork boundaries. Workers used apps and other mobile features to signal them when to stop working or if they had worked excessive overtime. Additionally, some workers created different browser profiles for work and nonwork web activities. The tactic least implemented in this sample of remote workers was communicative. Though few, some workers forewarned potential boundary violators that they should not be disturbed during work times (*Setting expectations*).

Three additional strategies emerged in our sample of remote workers that were not included in Kreiner et al. (2009). Individuals reported emulating the routines and behaviors of the office environment to recreate the feeling of going to and from work and being at work (*Emulating office routine*). Some workers dressed in their normal work attire during work hours and even went outside when work ended, pretending that they were coming home from work. Individuals also reported being keen on disconnecting completely from work by turning off work phones and removing all work-related items

as they transitioned to nonwork hours (*Purposefully disconnecting*). Finally, although less common, individuals attempted to limit the opportunity for work and nonwork overlap by working during time when potential boundary violators (e.g., family members) were not awake (*Reducing work and home role overlap*). The two authors involved in coding discussed the newly identified subcategories and classified *purposefully disconnecting* and *reducing work and home role overlap* as temporal tactics and *emulating office routine* as a behavioral tactic.

DISCUSSION

The purpose of our study was to investigate the relationship between segmentation preferences and work-nonwork balance among a sample of individuals required to work remotely due to COVID-19. Because many individuals have been forced into remote work as a result of COVID-19, the current context offered the unique opportunity to investigate the predictive value of boundary management preferences on work-nonwork balance without the influence of self-selection effects.

Contrary to our first two hypotheses, we found that segmentation preferences were positively related to work-nonwork balance and that this relationship was consistent across a three-month period. There are several possible explanations for our unexpected results. Our sample was comprised of workers who did not work remotely on a regular basis prior to the pandemic. It may be that our participants were thus recruited from a sample already predisposed toward the desire for segmentation. That is, our sample may have been nonrandom in the sense that it primarily included individuals who prefer segmentation. Indeed, our mean level of segmentation preferences was high and the standard deviation was low relative to the means and standard deviations observed in other studies not restricted to those with little prior remote work experience (e.g., Kreiner, 2006; Shockley & Allen, 2010). In addition, the timing of our study was such that at the time of our initial data collection individuals had already been working remotely for over a month. Thus, the higher than usual mean and compressed variation for segmentation could be based on expressed need due to the current context. While speculative without an assessment of boundary management preferences prior to COVID-19, it could be that the desire to segment had been enhanced from the experience of working remotely.

Our focus on work-nonwork rather than on work-family may have also been a factor in our findings. Our sample differs from samples typically used in work-family research. Work-family studies often restrict the sample to those who are partnered and/or have dependents who need caregiving in the home. Without such restrictions, our sample is younger, less likely

to be married, and less likely to have children than commonly found in work-family research. Moreover, the reduction in time spent commuting may free up time that can be devoted to nonwork activities. In addition, it may be that integration situations such as remote work are more conducive to aspects of nonwork that do not necessarily involve family, such as engagement in exercise, hobbies, and other forms of leisure. It may be that work-nonwork balance lends itself more to integration contexts than does work-family balance.

Contrary to our second two hypotheses, we also did not find support for our moderation hypotheses. Neither a dedicated home office nor the number of individuals in the household modified the relationship between segmentation preferences and work-nonwork balance. However, we did find significant direct relationships between the hypothesized moderators and work-nonwork balance. Specifically, having a dedicated home office space was positively related to work-nonwork balance. Our finding provides empirical backing for the advice often given to remote workers to establish a dedicated workspace within the home (e.g., Shockley & Clark, 2020). Moreover, further underscoring the importance of the home environment, our drop-out analyses indicated that having a dedicated home office space may have influenced individuals' likelihood of continuing to work from home during the COVID-19 pandemic. Such a space appears to be beneficial regardless of one's boundary management preferences.

We also found that having more individuals within the home was associated with less work-nonwork balance. This finding is consistent with research that shows a negative relationship between family demands and work-family conflict (e.g., Allen et al., 2020). Our findings demonstrate that an increased number of housemates may hinder work-nonwork balance, regardless of one's boundary management preferences. Work-family studies typically focus on number and/or age of children in the home. Our findings suggest the importance of including number of members in the household and consideration of a number of different living configurations when examining work-nonwork experiences. In future studies it would be interesting to investigate the boundary management preferences of household members. For example, two partners who work from home who share the same boundary management preferences may find it easier to effectively co-work from home than two partners with different preferences. This could extend to coworkers in office environments as well.

Our qualitative research reveals the specific strategies remote workers use to segment work and nonwork. Our intent was primarily to reveal the ways individuals segmented work-nonwork boundaries within the remote work context, rather than to what degree they used various known strategies; however, several strategies stood out. Overall, strategies that involved temporal

tactics were the most frequently reported. This included several new subcategories that emerged beyond those identified by Kreiner et al. (2009) that were more specific to the work from home environment. As noted by Cho (2020), the pandemic may necessitate new boundary management strategies to compensate for blurred physical role boundaries. For example, once the workday was completed individuals strategically disconnected and/or stored tools connected to work. This effort to disconnect is consistent with the recovery literature, which highlights the importance of being able to physically and mentally disengage from work in order for recovery to occur (Sonnentag & Fritz, 2007). Consistent with our finding with regard to the benefit of having a dedicated office, one of the most common strategies participants reported involved manipulating physical space. For example, individuals reported designating a specific space in the home for work and not taking work-related items into nonwork spaces. Again, such strategies may aid in the recovery process. Sonnentag, Kuttler, and Fritz (2010) found that spatial boundaries were associated with greater psychological detachment. Our findings underscore the importance of being able to create a physical border that separates work from nonwork. Overall, our findings suggested that many individuals attempted to mimic the rhythm and flow of work associated with “going to work” outside of the home. Future research that looks at how individuals construct and plan their remote workdays may yield additional insight into boundary management among remote workers.

Limitations

Our research has several limitations that should be noted. One is that the data are all self-report. This mode of data collection is consistent with our research questions and the inherently subjective topics of focus. However, it would be interesting in future studies to collect reports of boundary management strategies as assessed by coworkers and household members of the focal employee. Our use of panel data permitted us to sample individuals across a wide array of industries and occupations, and also raises concerns regarding self-selection and generalizability. For example, individuals struggling with work-nonwork balance concerns may have been less likely to take the time to participate in the study. Moreover, we cannot say to what extent our findings would generalize beyond the current pandemic situation. This is a particular concern given our unexpected results. We cannot say with certainty if these results are simply a product of the current context or if they are relevant to remote work in general. Similarly, the boundary management strategies that were reported as in use in the current study are bounded by the pandemic as well as the home context of this specific sample.

Implications

Our research has several theoretical and applied implications. We extend the work on boundary management preferences beyond work-family conflict to that of work-nonwork balance. Existing research has found little evidence of a direct relationship between boundary management preferences and work-family conflict. Our findings show a positive relationship between segmentation preferences and work-nonwork balance under a situational context (i.e., nonvoluntarily initiated remote work) that would seemingly lend itself to misalignment among those who prefer segmentation. Under the strong situation of forced remote work, preference for segmentation seemingly enabled individuals to balance their work and nonwork roles. Boundary management theory advancement may come from incorporation of a more dynamic perspective that considers the impact of shock events or other life events (e.g., birth of a baby) on preferences and behaviors. We have little knowledge with regard to how preferences and strategies may evolve across the lifespan or in response to external forces. Future research is needed to examine how boundary management preferences may change in response to work-nonwork shock events.

In addition, a *demands-abilities* fit approach may be valuable theoretical extension to the boundary management literature. Demands-abilities fit refers to the match between an individual's abilities and the demands of the environment. The positive relationship between segmentation preferences and work-nonwork balance we detected could suggest that individuals higher in segmentation preferences were more readily able to adjust to remote work because they had previously developed skills and strategies associated with segmentation prior to COVID-19. This would suggest viewing segmentation-integration behaviors as a skill or an ability that can be developed to meet the demands of the situation. For example, there is some evidence that segmentation preferences are associated with psychological detachment from work (Park, Fritz, & Jex, 2011) and that greater recovery is associated with greater work-life balance (Wepfer et al., 2018). Thus, considering segmenting strategies as a tool for enabling recovery and work-nonwork balance may be useful. Such research could be a useful addition to the literature and consistent with emerging work by Kossek and colleagues that involves assessments of boundary management styles along with training intended to empower employees with regard to ways to self-regulate how, when, and where they can best work (Kossek, 2016; Kossek, Ruderman, Braddy, & Hannum, 2012). Training interventions can be created to help individuals develop the abilities to match boundary management strategy with their work situations, including remote work (Kossek et al., 2012). This is important in that remote work appears positioned to stay well into the future as a growing number of

companies continue to extend their remote work plans (Mayer, 2020), making the development of such interventions tailored for the remote worker critical. Our qualitative data provide suggestions on how individuals who continue to work from home due to the pandemic and beyond may structure their work and nonwork roles in a way that meets their needs, but additional work is needed that provides guidance with regard to the effectiveness of various strategies.

Conclusion

Our study focused on boundary management among a sample of individuals who transitioned to remote work due to COVID-19. Several unexpected findings emerged, pointing to new needs for boundary management research and approaches. Given the likelihood of increased remote work well beyond the COVID-19 pandemic, a better understanding of boundary management processes for all types of remote workers will remain a research need for years to come.

REFERENCES

- Allen, T.D., Cho, E., & Meier, L.L. (2014). Work–family boundary dynamics. *Annual Review of Organizational Psychology and Organizational Behavior*, 1, 99–121.
- Allen, T.D., French, K.A., Dumani, S., & Shockley, K.M. (2020). A cross-national meta-analytic examination of predictors and outcomes associated with work–family conflict. *Journal of Applied Psychology*, 105, 539–576.
- Allen, T.D., Golden, T., & Shockley, K.M. (2015). How effective is telecommuting? Assessing the status of our scientific findings. *Psychological Science in the Public Interest*, 16(2), 40–68.
- Ashforth, B.E., Kreiner, G.E., & Fugate, M. (2000). All in a day's work: Boundaries and micro role transitions. *Academy of Management Review*, 25, 472–491.
- Brickman, P., Coates, D., & Janoff-Bulman, R. (1978). Lottery winners and accident victims: Is happiness relative? *Journal of Personality and Social Psychology*, 36, 917–927. <https://doi.org/10.1037/0022-3514.36.8.917>
- Bulger, C.A., Matthews, R.A., & Hoffman, M.E. (2007). Work and personal life boundary management: Boundary strength, work/personal life balance, and the segmentation-integration continuum. *Journal of Occupational Health Psychology*, 12(4), 365–375.
- Casper, W.J., Vaziri, H., Wayne, J.H., DeHauw, S., & Greenhaus, J. (2018). The jingle-jangle of work–nonwork balance: A comprehensive and meta-analytic review of its meaning and measurement. *Journal of Applied Psychology*, 103, 182–214.
- Chen, Z., Powell, G.N., & Greenhaus, J.H. (2009). Work-to-family conflict, positive spillover, and boundary management: A person–environment fit approach. *Journal of Vocational Behavior*, 74(1), 82–93. <https://doi.org/10.1016/j.jvb.2008.10.009>

- Cho, E. (2020). Examining boundaries to understand the impact of COVID-19 on vocational behaviors. *Journal of Vocational Behavior*, 119, <https://doi.org/10.1016/j.jvb.2020.103437>
- Craig, L., & Churchill, B. (2020). Dual-earner parent couples' work and care during COVID-19. *Gender, Work, & Organization*, 1–14. <https://doi.org/10.1111/gwao.12497>
- Desilver, D. (2020). *Working from home was a luxury for the relatively affluent before coronavirus - not any more*. World Economic Forum, March 21. Retrieved from <https://www.weforum.org/agenda/2020/03/working-from-home-coronavirus-workers-future-of-work/>
- Diener, E., Lucas, R.E., & Scollon, C.N. (2006). Beyond the hedonic treadmill: Revising the adaptation theory of well-being. *American Psychologist*, 61, 305–314. <https://doi.org/10.1037/0003-066X.61.4.305>
- Edwards, J.R., Cable, D.M., Williamson, I.O., Lambert, L.S., & Shipp, A.J. (2006). The phenomenology of fit: Linking the person and the environment to the subjective experience of person-environment fit. *Journal of Applied Psychology*, 91(802), 827.
- Edwards, J.R., & Rothbard, N.P. (1999). Work and family stress and well-being: An examination of person environment fit in the work and family domains. *Organizational Behavior and Human Decision Processes*, 77, 85–129.
- Fleiss, J.L. (1981). *Statistical methods for rates and proportions*. Hoboken, NJ: Wiley.
- Gajendran, R.S., Harrison, D.A., & Delaney-Klinger, K. (2015). Are telecommuters remotely good citizens? Unpacking telecommuting's effects on performance via i-deals and job resources. *Personnel Psychology*, 68(2), 353–393.
- Greenhaus, J., & Allen, T. (2011). Work-family balance: A review and extension of the literature. In J. Campbell Quick & L.E. Tetrick (Eds.), *Handbook of occupational health psychology* (2nd ed., pp. 156–183). Washington: American Psychological Association.
- Greenhaus, J.H., Ziegert, J.C., & Allen, T.D. (2012). When family-supportive supervision matters: Relations between multiple sources of support and work-family balance. *Journal of Vocational Behavior*, 80, 266–275.
- Koetsier, J. (2020). 58% of American knowledge workers are now working remotely. *Forbes*. Retrieved from <https://www.forbes.com/sites/johnkoetsier/2020/03/20/58-of-american-knowledge-workers-are-now-working-remotely/#1b21e5f93303>
- Kossek, E.E. (2016). Implementing organizational work-life interventions: Toward a triple bottom line. *Community, Work & Family*, 19, 242–256.
- Kossek, E.E., & Lautsch, B. (2012). Work-family boundary management styles in organizations: A cross-level model. *Organizational Psychology Review*, 2(2), 152–171. <https://doi.org/10.1177/2041386611436264>
- Kossek, E.E., Lautsch, B.A., & Eaton, S.C. (2006). Telecommuting, control, and boundary management: Correlates of policy use and practice, job control, and work-family effectiveness. *Journal of Vocational Behavior*, 68(2), 347–367.
- Kossek, E., Ruderman, M., Braddy, P., & Hannum, K. (2012). Work-nonwork boundary management profiles: A person-centered approach. *Journal of Vocational Behavior*, 81, 112–128. <https://doi.org/10.1016/j.jvb.2012.04.003>

- Kreiner, G.E. (2006). Consequences of work-home segmentation or integration: A person-environment fit perspective. *Journal of Organizational Behavior*, 27(4), 485–507.
- Kreiner, G.E., Hollensbe, E.C., & Sheep, M.L. (2009). Balancing borders and bridges: Negotiating the work-home interface via boundary work tactics. *Academy of Management Journal*, 52, 704–730.
- Kristof, A.L. (1996). Person–organization fit: An integrative review of its conceptualization, measurement, and implications. *Personnel Psychology*, 49, 1–49.
- Lapierre, L.M., Van Steenbergen, E.F., Peeters, M.C., & Kluwer, E.S. (2016). Juggling work and family responsibilities when involuntarily working more from home: A multiwave study of financial sales professionals. *Journal of Organizational Behavior*, 37(6), 804–822.
- Luhmann, M., Hofmann, W., Eid, M., & Lucas, R.E. (2012). Subjective well-being and adaptation to life events: A meta-analysis on differences between cognitive and affective well-being. *Journal of Personality and Social Psychology*, 120, 592–615.
- Martin, B.H., & MacDonnell, R. (2012). Is telework effective for organizations? A meta-analysis of empirical research on perceptions of telework and organizational outcomes. *Management Research Review*, 35(7), 602–616.
- Mayer, K. (2020, August 10). Why embracing long-term remote work is the right thing to do. *Human Resource Executive*. Retrieved from https://siop.caboodleai.net/en/article/17186?utm_source=ElasticEmail-sb-siop&utm_medium=newsletter&utm_campaign=siop-783-s-en-120820&itd_did=783&itd_uid=2316
- Nippert-Eng, C.E. (1996). Calendars and keys: The classification of “home” and “work”. *Sociological Forum*, 11, 563–582.
- Olson-Buchanan, J.B., Boswell, W.R., & Morgan, T.J. (2016). The role of technology in managing the work and nonwork interface. In T.D. Allen & L.T. Eby (Eds.), *The Oxford handbook of work and family* (pp. 333–348). Oxford, UK: Oxford University Press.
- Park, Y., Fritz, C., & Jex, S.M. (2011). Relationships between work-home segmentation and psychological detachment from work: The role of communication technology use at home. *Journal of Occupational Health Psychology*, 16, 457–467.
- Powell, G.N., Greenhaus, J.H., Allen, T.D., & Johnson, R.E. (2019). Introduction to special topic forum: Advancing and expanding work-life theory from multiple perspectives. *Academy of Management Review*, 44(1), 54–71.
- Randolph, J.J. (2008). *Online Kappa calculator* [Computer software]. Retrieved from <http://justus.randolph.name/kappa>
- Rothausen, T.J. (2011). Unpacking work-family: Core overarching but under-identified issues. *Industrial and Organizational Psychology: Perspectives on Science and Practice*, 4, 379–384.
- Rothbard, N.P., Phillips, K.W., & Dumas, T.L. (2005). Managing multiple roles: Work-family policies and individuals’ desires for segmentation. *Organization Science*, 16(3), 243–258.
- Shockley, K.M., & Allen, T.D. (2010). Investigating the missing link in flexible work arrangement utilization: An individual difference perspective. *Journal of Vocational Behavior*, 76(1), 131–142.

- Shockley, K.M. Clark M. (2020). *Work-family balance struggles in the time of COVID-19*. Bowling Green, OH: Society for Industrial and Organizational Psychology. Retrieved from <https://www.sio.org/Research-Publications/Items-of-Interest/ArtMID/19366/ArticleID/3463/Work-Family-Balance-Struggles-in-the-Time-of-COVID-19>
- Shrout, P.E., & Lane, S.P. (2012). Psychometrics. In M.R. Mehl & T.A. Conner (Eds.) *Handbook of research methods for studying daily life* (pp. 302–320). New York, NY: Guilford Press.
- Sonnentag, S., & Fritz, C. (2007). The recovery experience questionnaire: Development and validation of a measure for assessing recuperation and unwinding from work. *Journal of Occupational Health Psychology*, 12(3), 204–221. <https://doi.org/10.1037/1076-8998.12.3.204>
- Sonnentag, S., Kuttler, I., & Fritz, C. (2010). Job stressors, emotional exhaustion, and need for recovery: A multi-source study on the benefits of psychological detachment from work during non-work time. *Journal of Vocational Behavior*, 76, 355–365.
- US Bureau of Labor Statistics (2018). *Economic news release*. Washington: U.S. Bureau of Labor Statistics. Retrieved from <https://www.bls.gov/news.release/atus.t06.htm>
- Vaziri, H., Casper, W.J., Wayne, J.H., & Matthews, R.A. (2020). Changes to the work–family interface during the COVID-19 pandemic: Examining predictors and implications using latent transition analysis. *Journal of Applied Psychology*, 105(10), 1073–1087. <https://doi.org/10.1037/apl0000819>
- Wepfer, A.G., Allen, T.D., Brauchli, R., Jenny, G.J., & Bauer, G.F. (2018). Work-life boundaries and well-being: Does work-to-life integration impair well-being through lack of recovery? *Journal of Business and Psychology*, 33(6), 727–740.